CODE N' CREATE | LED CIRCUIT CHALLENGES

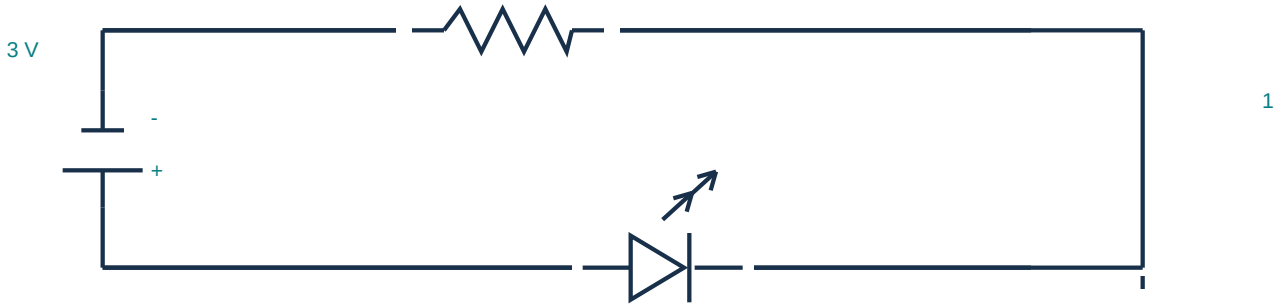
Tuesday Lab | Grades 9-10 | Name: _____

Equipment: 2 AA cells (3 V), 5 mm LEDs, 4-band resistors, multimeter, leads

SAFETY: Never power an LED without a current-limiting resistor.
Power OFF when changing wiring or measuring resistance. Teacher checks each build.

Symbols: zigzag = resistor; triangle-like diode with outward arrows = LED; long/short lines = battery.

Challenge 1 - One resistor, one LED

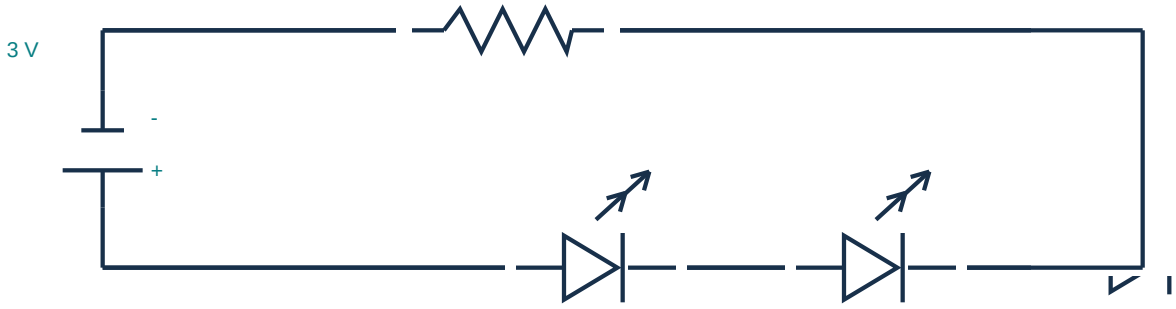


1. Record resistor color bands and calculated value: _____

2. Predict and record LED brightness: _____

3. What happens if you reverse the LED? Why? _____

Challenge 2 - One resistor, two LEDs in series



4. Predict: Will both LEDs light with 3 V? Record what actually happens.

5. Compare brightness with Challenge 1. Explain your observation.

CODE N' CREATE | CIRCUIT INVESTIGATIONS

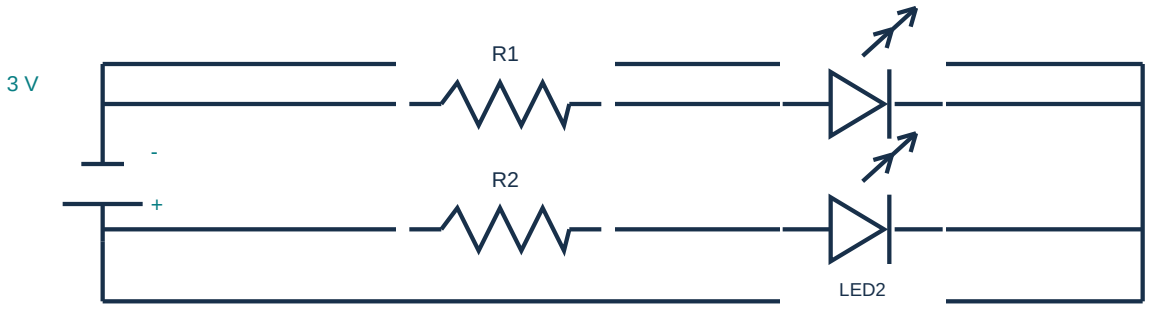
Grades 9-10 | Name: _____

Safety investigation - LED without a resistor (DO NOT BUILD)

Predict what could happen if a 3 V battery is connected directly to a 5 mm LED.

Explain how a resistor protects an LED: _____

Challenge 3 - Two parallel branches (R + LED in each)



6. Which branches light? Compare LED brightness and explain.

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7. Disconnect one branch with power OFF. Predict what happens to the other.

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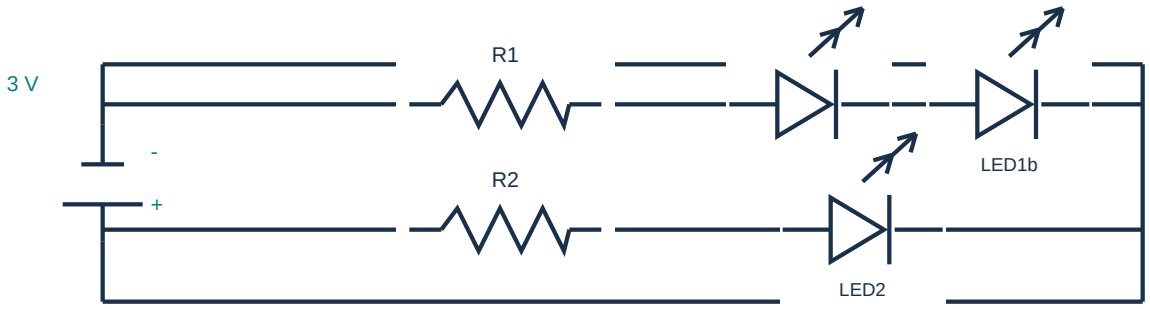
8. Measure each resistor separately with power OFF: R1 ____ ohms; R2 ____ ohms.

CODE N' CREATE | ADVANCED MIXED-BRANCH CIRCUIT

Grades 9-10 | Name: _____

Challenge 4 - Branch A: R + two LEDs; Branch B: R + one LED



9. Predict which branch will light at 3 V. Explain using LED forward voltage.

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10. Record the observed behavior of each branch. Was your prediction correct?

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11. Why does each branch need its own current-limiting resistor?

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12. How would increasing supply voltage change the result? Do NOT increase it today.

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13. Measure R1 and R2 separately with power OFF. Record units and tolerance.

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14. Explain why this is a parallel circuit containing series components.

Optional calculation (assume a red LED has ~2 V forward drop):

For a 3 V supply and 220 ohm resistor, estimate single-LED branch current.

Teacher note: Two series red LEDs may not turn on at 3 V.